

Exploring the Relationship Between High School Students' Physics-Related Personal Epistemologies and Self-regulated Learning in Turkey

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Abstract This article reports on an empirical exploration of the relations and strengths among Turkish grades 9–11 students' ($n = 209$) personal epistemologies (justification of knowledge, certainty of knowledge, source of knowledge, development of knowledge), self-regulated learning (extrinsic motivation, intrinsic motivation, rehearsal, elaboration, organization, critical thinking, metacognitive self-regulation), and achievement in physics (course grades). Established instruments were used to collect data on these students' beliefs about knowledge and components of self-regulated learning (SRL) such as goal orientations (extrinsic and intrinsic motivation) and learning strategies, critical thinking, and metacognitive regulation. Results from structural equation modeling revealed that students' personal epistemologies directly influenced their motivation (extrinsic and intrinsic goal orientations), rehearsal and organization strategies, and metacognitive self-regulation to learn physics. Furthermore, students' personal epistemologies indirectly (mediated through motivation beliefs) influenced rehearsal, elaboration and organization strategies, critical thinking, and metacognitive self-regulation to learn physics. Students' ideas about knowledge and knowing about the source and development of knowledge significantly contributed to students' self-regulatory skills and physics course grade. Implications and future directions are discussed.

Keywords Personal epistemology · Physics education · Self-regulated learning · Structural equation modeling

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Introduction

Views about learning science have changed over the last 50+ years from simple conditioning to complex networks of interacting factors, moving from behaviorism as the dominant view in the 1950s to cognitive development in the 1960s, constructivism in the 1980s, and cognitive science in the 1990s. Collectively, the strengths and weaknesses of these views have partially influenced the twenty-first century transition in viewing science learning as a *hot* rather than a *cold* process where the learner's personal affective attributes are considered along with the somewhat sterile cognition factors. Epistemology and self-regulated learning (SRL) have potential in clarifying this transition. Personal epistemologies refer to students' ideas about knowledge and knowing (Hofer, 2001), while SRL refers to a process in which students actively regulate their own motivation, cognition, and behavior toward the successful completion of academic tasks (Winne & Hadwin, 1998). Both personal epistemology and SRL play important roles in students' learning in general, but little is known about their roles in learning high school physics. Recently, some researchers claimed that the influence of personal epistemology on learning and achievement is more likely to be indirect rather than direct (Muis, 2007; Schommer-Aikins, Duell & Hutter, 2005). Furthermore, Hofer and Sinatra (2010) emphasized a need for further studies to describe and model the complex interplays between personal epistemologies and SRL on physics teaching and learning. Therefore, it is assumed that better understanding of the underlying relations in the model for learning physics will help guide the development or more robust curriculum standards and teaching methods.

Research on personal epistemology and SRL is particularly important for the efforts to promote students' understanding of the nature of science (NOS) and the practices and big ideas of science. Personal epistemology highlights some essential characteristics of NOS, including the structure of scientific knowledge, tentativeness, evidence-based authority, and justification. Studies have explored the relationships of personal epistemology with strategies for learning science (Bromme, Pieschl & Stahl, 2010) and the decision-making process (Kardash & Scholes, 1996). SRL is assumed to be an important determinant of learning and academic performance (Hsu, Yen, Chang, Wang & Chen, 2014). Self-regulated learners in science education are those who (a) tend to view learning science as intrinsically interesting and (b) control their own process toward self-set goals and effectiveness of their learning approaches (Credé & Phillips, 2011). Hence, it is important to help students foster self-regulatory skills that allow them to control their learning through productive motivational beliefs and use of metacognitive learning strategies.

Personal epistemology is viewed as a starting point for learning about the nature and development of science knowledge in the classroom (National Research Council [NRC], 2007). Students' ideas about knowledge and knowing appear to guide their actions toward the acquisition of knowledge (Hofer, 2001; Muis, 2007). Therefore, students' personal epistemologies may relate to their SRL in physics. Studies concerning personal epistemology and SRL have most often been conducted at the university level; high school students have been the subject of few studies (Muis, Bendixen & Haerle, 2006). Furthermore, domain specificity is viewed as another factor that influences students' personal epistemologies (Muis et al., 2006; Topçu, 2013) and SRL (Pintrich, 2000). Understanding the functions of discipline-specific personal

epistemologies in SRL may help educators to more effectively teach physics concepts to high school students. Therefore, there is a need to determine the interplays and structural relationships between the personal epistemologies and SRL with high school students who are just starting to develop ideas and SRL strategies in physics (Muis et al., 2006).

Personal Epistemology and Achievement

Personal epistemology refers to individuals' ideas about what counts as knowledge and how knowledge is constructed (Hofer, 2001). Hofer and Pintrich (2002) theorized that personal epistemology was based on four interrelated dimensions about knowledge and learning. The first two dimensions relate to the nature of knowledge, and the latter two describe the process of knowing: (a) *certainty of knowledge* refers to the strength of supporting evidence and the tentativeness of knowledge, (b) *development of knowledge* focuses on the connectedness of knowledge, (c) *justification of knowledge* focuses on how individuals evaluate knowledge claims, and (d) *source of knowledge* indicates that knowledge is constructed by external sources or by learners. Each dimension would likely reflect the nature of the discipline to be learned, which might differ drastically between the arts, humanities, and sciences.

It has been reported that personal epistemologies are correlated to students' skill and attitude for learning science. Some researchers have suggested that personal epistemology directly and indirectly predicts students' achievement in science (Koksal, 2011; Taasooobshirazi & Sinatra, 2011). Stathopoulou and Vosniadou (2007) studied the influence of personal epistemology on achievement in classical dynamics and found that grade 10 students in Greece with low- and high-epistemological sophistication had significant ($p < .001$) differences in physics achievement; the sophisticated group achieved higher scores than the low (naïve) group. Naïve beliefs regarding the certainty and development of knowledge have been found to be negatively correlated with the skill to interpret controversial evidence (Kardash & Scholes, 1996).

Recently, researchers have claimed that students' personal epistemologies may be different across disciplines (Muis et al., 2006). Topçu (2013), for example, examined Turkish preservice teachers' personal epistemologies in chemistry, biology, and physics and found that these university students hold different personal epistemologies across these science disciplines in all dimensions of personal epistemology. Students reported that they viewed knowledge in physics as more certain than chemistry and biology. He concluded that students viewed biology as personal evaluation and physics as authority; he suggested further research is needed with different contexts and younger students to more fully explore the discipline-specific nature of personal epistemologies.

Self-regulated Learning and Achievement

SRL theories explain how motivational, cognition, and contextual factors influence the learning process while self-regulated learners actively and constructively monitor their learning (Pintrich, 2000; Winne & Hadwin, 1998). Winne and Hadwin (1998) theorized in their COPES (i.e. conditions, operations, products, evaluations, standards) model of SRL that learning occurs in four basic phases: task definition, goal setting and planning, enactment, and adaptation to metacognition. While in the task definition stage, the

student generates perceptions of what the task is; in the goal-setting and planning stage, the student generates goals for the task based on the task definition; in the enactment phase, the student carries out the plan of study and tactics created in the goal-setting phase; in the adaptation phase, the student may show various types of reflections and adaptations about the learning event based on experience of the task. Winne and Hadwin described each of these four phases in terms of the interaction of a person's conditions, operations, products, evaluations, and standards.

Research has documented the importance of students' SRL on achievement in science (Hsu et al., 2014; NRC, 2007; Taasobshirazi & Sinatra, 2011). Students with high SRL demonstrated higher levels of involvement and effort on academic tasks than those with low SRL (Eilam, Zeidner & Aharon, 2009). As a part of SRL, metacognition has "an interface–retrieval–transportation role between knowledge construction, prior knowledge in long-term memory, and sensory input in short-term memory" (Yore & Treagust, 2006, p. 294). Hsu et al.'s (2014) review of studies in science reading using a SRL lens revealed that metacognitive guidance was the most effective intervention to promote science reading. Therefore, it is important to examine the relationship of SRL with other variables to promote scientific literacy (Yore & Treagust, 2006) and achievement in physics (Taasobshirazi & Sinatra, 2011).

Goal orientation dimensions (e.g. intrinsic goal orientation) of SRL are essential to lead the students to use learning strategies effectively (Koksal, 2011). Intrinsically goal-oriented students, for example, are more highly motivated to learn and use deeper cognition strategies (Taasobshirazi & Sinatra, 2011). Gürçay and Balta (2013) reported a significant ($p < .05$) relationship between intrinsic goal orientation and metacognitive self-regulation that explained 28 % of the variance in self-regulation in physics.

Metacognitive students actively decide how to use resources effectively and make judgments about outcomes and learning (Pintrich, 2000). Yumusak, Sungur and Cakiroglu (2007) explored the contribution of SRL to Turkish high school students' achievement in biology using the Turkish version of the Motivated Strategies for Learning Questionnaire (MSLQ; Sungur, 2004). They found that grade 10 students' ($n = 214$) motivational beliefs explained 10 % of the variation in their achievement. Extrinsic goal orientation and task value significantly predicted student achievement ($p < .005$) while other dimensions did not. Cognitive and metacognitive strategies were significantly ($p < .05$) related to achievement and accounted for 9 % of the variation.

Personal Epistemology, Self-regulated Learning, and Achievement

Research has reported that students' personal epistemologies are correlated to their SRL strategies (Hofer, 2001; Muis & Franco, 2009). Personal epistemology can affect thinking in many ways, for example, directing one's perception and attention to particular features of information (Hofer, 2001). Hofer (2001) stated that students' personal epistemologies relate to "the goals and standards that determine engagement in learning, depth of processing, and comprehension monitoring" (p. 370). A learner who has a naïve view about the source and justification of knowledge uses learning strategies that are different from a learner with a sophisticated view. A naïve learner relies on only one source (e.g. a textbook); a sophisticated learner tends to

look for different sources, monitors epistemic claims, weighs evidence, and evaluates authorities.

Muis (2007) proposed a theoretical model to describe how personal epistemology facilitates or limits facets of SRL. Her model assumes the relationship between personal epistemology and SRL is reciprocal and discipline specific. Based on Winne and Hadwin's (1998) and Pintrich's (2000) models on SRL, Muis specified four propositions corresponding to the four phases of SRL that explain the relationship between personal epistemology and SRL.

Phase 1—Personal epistemology is one component of the cognition and affective conditions on task definition. Students activate their perceptions of the task, context, and knowledge of the task, thereby helping to define the conditions of the task. Muis (2007) suggested that personal epistemology is a component of cognition and affective conditions and a key element to task definition and forethought.

Phase 2—Personal epistemology influences goal standards students set. Students set goals to pursue by a particular strategy or a set of strategies. Muis (2007) proposed that students' personal epistemologies facilitate or constrain facets of SRL by relating them to the goals for the task. She asserted that personal epistemology might function as implicit theories that can induce particular types of goals for learning, including mastery or performance-oriented goals.

Phase 3—Personal epistemology translates into epistemic standards that serve as inputs to metacognition. Students understand and predict the complexity of the problem, the certainty of knowledge, and how they evaluate the evidence (Kuhn, 2000). Hofer (2004) suggested that personal epistemology relates to the metacognition process for any task by influencing epistemological standards that the student sets for that learning task. These standards serve as pertinent information during metacognitive monitoring (Muis, 2007).

Phase 4—SRL may play a role in the development of personal epistemology. Students use reciprocal feedback loops to inform their SRL and personal epistemology. Theorists agree that SRL is a cyclical construct in which information produced in any phase can feed into the same phase or other phases (e.g. Winne & Hadwin, 1998; Pintrich, 2000). Muis (2007) posited that the reciprocal, bidirectional relationship is an essential feature of the model to illustrate that information from any phase can inform other components.

Muis and Franco (2009) explored educational psychology students ($n = 201$) at a Canadian university to empirically test Muis' (2007) hypothesis. They conducted a structural equation model of data from three instruments—Discipline-Focused Epistemological Beliefs Questionnaire (Hofer, 2000), Achievement Goals Questionnaire (Elliot & McGregor, 2001), and a short version of the MSLQ (Pintrich, Smith, Garcia & McKeachie, 1991)—and the students' final course grade as the achievement score. SEM revealed that beliefs on certainty of knowledge predicted extrinsic goal orientation ($\beta = 0.29$) and that students' goal orientations were significantly related to students' SRL strategies; intrinsic goal orientation predicted the use of rehearsal ($\beta = 0.82$), critical thinking ($\beta = 0.72$), and metacognitive self-regulation ($\beta = 0.97$). Furthermore, SRL strategies (i.e. rehearsal, metacognitive self-

regulation, and critical thinking) significantly predicted achievement ($\beta = 0.14$, $\beta = 0.69$, and $\beta = 0.29$, respectively).

The Current Study

The purpose of this study was to use structural equation modeling (SEM) to examine the relationships among personal epistemology, SRL, and high school physics achievement. The Muis (2007) theoretical model (Fig. 1) was used to focus the SEM. The solid lines in the model specify the direct relationships, and the dashed lines display the indirect relationships.

The first direct relationship specifies that students' ideas about knowledge and knowing in physics are related to goal orientation dimensions of SRL including extrinsic and intrinsic goal orientations toward physics. The second direct relationship specifies that students' goal orientations are related to their use of cognitive and metacognitive strategies including rehearsal, organization, critical thinking, elaboration, and metacognitive self-regulation in physics. The third direct relationship among strategies and physics achievement involves the use of cognitive and metacognitive strategies to construct knowledge about physics. The first indirect relationship in the model specifies that students' ideas about knowledge and knowing are indirectly related to their use of cognitive and metacognitive strategies through the moderation of goal orientation. The second indirect relationship is about the indirect effect of personal epistemology on students' achievement in physics through the moderation of goal orientation and cognitive and metacognitive strategies. The third indirect relationship is about the effect of goal orientation on students' achievement in physics involving the mediation of cognitive and metacognitive strategies. The fourth proposition (i.e. SRL may influence personal epistemology) in Muis' (2007) model was not tested because it requires a time-lag and repeated data collection that was not possible in this study.

It has been suggested that studies of epistemology in science education should pay attention to domain specificity and analysis of complex interplays among personal epistemology and learning approaches (Yang & Tsai, 2012). Physics is seen by students as an abstract and certain discipline (Topçu, 2013); therefore, it is important to examine the relations between personal epistemology and SRL in physics. Moreover, we have not located any study addressing the relations between personal epistemology and SRL in physics for the Turkish context. This study sought to address the following research questions:

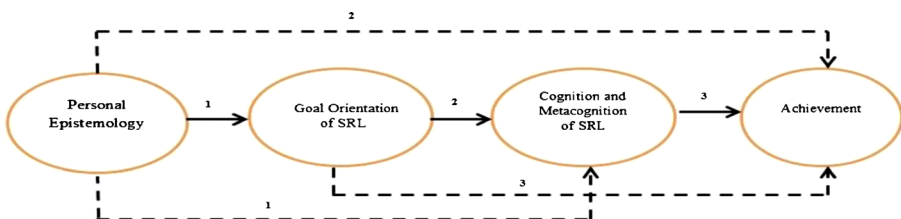


Fig. 1 Adapted model of personal epistemology and self-regulated learning (Muis, 2007, p. 177)

1. What proportion of variance in physics achievement is explained by physics-related personal epistemology and SRL in Turkish high school students?
2. What extent do the dimensions of personal epistemology express the goal orientation that Turkish students use in physics?
3. What extent do the dimensions of personal epistemology predict the cognitive and metacognitive strategies that Turkish students use in physics through the mediation of goal orientation?
4. What extent does the hypothesized model fit the SEM of empirical data obtained from Turkish students?

Methods

The research methodology of this study is a quantitative survey design to address the research questions. Established self-report questionnaires and classroom assessments were used to collect data. The SEM approach was employed to explore the relationships among the variables.

Context and Potential Participants

Turkey has a centralized educational system where the elementary and high school curricula are developed and regulated by the Ministry of National Education (MoNE). The national physics curriculum expects students to understand the interrelationships between physics, technology, society, and environment, to develop problem-solving skills, and to understand the basic nature of physics knowledge (MoNE, 2011). The students are also expected to adopt values and attitudes toward physics and be able to critically analyze and reflect on physics-related scientific and technological issues in their socioeconomic environment (MoNE, 2011). Starting in the 2013–2014 academic year, physics is a compulsory course for grades 9 and 10 (2 h a week) and an elective course for grades 11 and 12 (2–3 h a week). During the semester that this study took place, physics was compulsory for only grade 9 and elective for grades 10–12; therefore, student characteristics will vary between the grade levels. The national physics curriculum included the following topics: grade 9—nature of physics, matter, force and motion, energy, electricity and magnetisms, and waves; grade 10—force and motion, electricity, modern physics, and waves; grade 11—matter, force and motion, magnetisms, modern physics, waves, and astronomy; and grade 12—matter, force and motion, electricity and electronics, waves, modern physics, quarks in atoms, and the nature of physics.

Convenience sampling was used in this study (Creswell, 2007). Bursa, a metropolitan city located in northwestern Anatolia, was chosen because of its convenience to the researcher. Bursa is the fourth most populous (2.7 million people in 2013) and most industrialized metropolitan center in Turkey. All high school students in Bursa Province were identified as the target population. However, the Office of the Turkish Ministry of Education in Bursa allocated and identified only one public high school for data collection in this study. Therefore, the students from this school were identified as the participants. The school was located in the city center and had 780 students who were moderate achievers and socioeconomically diverse.

Instruments

The questionnaires, described more fully below, were selected for this study because (a) they are adaptable to the domain of physics, (b) they captured the facets of the two constructs (personal epistemology and SRL) in the proposed model, and (c) they have been translated and validated in Turkey.

Modified Epistemic Beliefs Questionnaire

The original Modified Epistemic Beliefs Questionnaire (MEBQ; Conley, Pintrich, Wekiri & Harrison, 2004; Hofer & Pintrich, 2002), a five-point Likert-type scale, consisted of 26 items to measure students' personal epistemology in four dimensions: certainty of knowledge (five items), source of knowledge (four items), justification of knowledge (nine items), and development of knowledge (six items). Items in the certainty and source dimensions were reversed so that higher scores represented more sophisticated beliefs. Ozkan (2008) adapted the questionnaire items from English into Turkish using the established translation–back translation technique. The Turkish version of the questionnaire was assumed to be valid based on the original validation and the rigorous translation technique. Kurt (2009) explored the MEBQ's reliability with 1557 middle and high school students and found the following reliabilities: 0.59 for source, 0.59 for certainty, 0.83 for justification, and 0.61 for development. Since the purpose of the present study is to identify students' personal epistemologies in physics, we replaced the words “science” and “scientists” with “physics” and “physicists.”

The confirmatory factor analysis (CFA) for the MEBQ resulted in χ^2 (290, $N = 209$) = 658.77, $p < .001$, standardized root mean square residual (SRMR) = 0.07, root mean square error of approximation (RMSEA) = 0.08, and comparative fit index (CFI) = 0.80. These results indicated that the model did not fit with the expected level. Cabrera-Nguyen (2010) suggested using absolute cutoff values of 0.30 for factor loading. This step supported the deletion of two items (i.e. one each from the certainty and source dimensions) with factor loadings of 0.25 and 0.28. The CFA was re-run and those results indicated a moderate model fit, χ^2 (243, $N = 209$) = 407.30, $p < .001$, SRMR = 0.06, RMSEA = 0.06, CFI = 0.90. Cronbach's alphas for reliability were 0.67 for source, 0.80 for certainty, 0.85 for justification, and 0.78 for development. The CFA and internal consistency results indicated that the Turkish version of the MEBQ was valid and reliable.

Motivated Strategies for Learning Questionnaire

The MSLQ, a seven-point Likert-type self-report instrument (Pintrich et al., 1991), was designed to measure students' goals and strategies in any domain. The MSLQ has been translated into and adapted for Turkish using appropriate translation techniques (Sungur, 2004) and used by other researchers (Yumusak et al., 2007). A short version of the MSLQ used in this study had corresponding scales to those in the proposed model and are reported elsewhere (Muis, 2007; Muis & Franco, 2009). The MSLQ scales were intrinsic goal orientation—assessed students' views of curiosity and

mastery (4 items), extrinsic goal orientation—assessed students' views of grades and rewards (4 items), rehearsal—assessed how often students used strategies involving reciting or memorizing (4 items), elaboration—assessed how often students used strategies helping to build internal connections between items (6 items), organization—assessed how often students used strategies that help select appropriate information and construct understanding among the information (4 items), critical thinking—assessed what degree students report applying previous knowledge to new situations in order to make critical decisions (5 items) and metacognitive self-regulation—assessed what degree students report awareness, knowledge, and control of cognition (12 items). Because the purpose of the present study is to identify students' SRL strategies in physics, the items were adapted to physics by changing “biology” words with “physics” words (Lin, Liang & Tsai, 2012). Two items in the metacognitive self-regulation scale were negatively worded, so these item responses were reversed ($1 = 7, \dots, 7 = 1$) before a score was computed.

The validity of the Turkish MSLQ was assumed from the validation of the original instrument; the reliability of the Turkish version was explored by using responses from the 519 high school students in the Yumusak et al. (2007) study. They reported the reliability of the MSLQ subscales as 0.64 for intrinsic goal orientation, 0.54 for extrinsic goal orientation, 0.66 for rehearsal, 0.75 for elaboration, 0.68 for organization, 0.78 for critical thinking, and 0.77 for metacognitive self-regulation. The CFA results were a moderate model fit, χ^2 (662, $N = 209$) = 1349.86, $p < .0005$, SRMR = 0.06, RMSEA = 0.06, CFI = 0.91. Cronbach's alpha values were 0.68 for intrinsic goal orientation, 0.74 for extrinsic goal orientation, 0.71 for rehearsal, 0.83 for elaboration, 0.74 for organization, 0.84 for critical thinking, and 0.84 for metacognitive self-regulation, which indicate an acceptable reliability.

Achievement in Physics

The students' final physics grade in the semester that the study took place was used as their achievement score because an external test would take extra time of participants, which was beyond the time demands reasonable for the school, teachers, and participants. Using students' school grade might be problematic due to a teacher's random assessment bias (Popham, 2012); however, these physics grades comprised the results of three teacher-made mid-term examinations that were equally weighted. Furthermore, all students at a given grade level were assessed by the same examination that the three physics teachers prepared together using the prescribed curriculum and learning outcomes. Therefore, the probability of unidirectional bias would be reduced. The average Cronbach's alpha for these examinations was 0.71, which was an acceptable reliability. The students' final course grades ranged from 1 (failed) to 5 (excellent), with a mean of 3.32 and a standard deviation of 1.10. Students' final grades were obtained from the school's student records.

Data Collection

Data were collected in May 2013. Students were introduced to the study by one of the researchers, who first explained the study's purpose and the students' rights as

participants. Students were invited to participate in the study and asked to take a parental permission form home for their parents to review and provide their consent. A week after the students returned the forms, classroom teachers gave the data collection instrument to those students who had their parental permission forms signed and volunteered to participate. The participants were given an hour to complete the instruments. A total of 209 (109 female, 100 male) students were involved in the study, 79 from grade 9, 57 from grade 10, and 73 from grade 11. There were no grade 12 participants because at the time of data collection they were taking the Undergraduate Placement Examination (UPE), which is required for admission to postsecondary education in Turkey.

Data Analysis

The SEM approach was utilized, using MPLUS 6 software, which allows the computation of direct and indirect effects among scales and provides estimates of all specific paths in the model (Kline, 2011). Furthermore, the SEM allows for overall test of the fit of a particular model to observed data. Maximum likelihood estimation procedure was used for the estimation of the model under consideration. Because the total number of participants was not large enough to construct each variable as a latent variable in the model (Kenny, 2014), each variable was considered as an observed variable. After the model was run, the model fit as well as the modification indices were examined. Based on the modification indices, the intercorrelations between scales were used if the theoretical model allowed.

Results

The purpose of this study was to investigate the relationship between physics-related personal epistemology and SRL, and how components of these two constructs could account for achievement in physics. Personal epistemology dimensions of source, certainty, justification, and development were measured using the MEBQ. SRL constructs including intrinsic motivation, extrinsic motivation, rehearsal, elaboration, organization, critical thinking, and metacognitive self-regulation were measured using the MSLQ. Academic performance was measured using students' final physics course grade.

Descriptive Statistics

The mean, standard deviation, skewness, and kurtosis value for each variable were calculated (Table 1). The descriptive statistics (mean scores and standard deviations) give insight about the students' personal epistemologies on a five-point scale. The students' personal epistemologies generally were between moderate and moderate–sophisticated with some variance; the means were close to, but above, the midpoint. The source of knowledge dimension had the highest mean value ($M = 3.84$), whereas the justification of knowledge dimension had the lowest mean value ($M = 3.21$).

The descriptive statistics (mean and standard deviations) of the students' motivational beliefs, and cognitive and metacognitive strategies measured on a

7-point scale (1 = low, 4 = moderate, 7 = high) indicated that their motivational beliefs toward physics and cognitive and metacognitive strategies were between moderate and moderate-high with some variation. Although extrinsic motivation had a higher mean value than intrinsic motivation, the difference between both values was quite small (~8 %). Among cognitive and metacognitive strategies, rehearsal strategy had the highest mean value of 5.24 ($SD = 1.13$) and metacognitive self-regulation strategy had the lowest mean value ($M = 4.01$, $SD = 1.86$). The difference among the MSLQ components was somewhat larger than the MEBQ components (~33 %). Physics achievement was measured on a five-point scale (1 = failed, 3 = satisfactory, 5 = excellent) indicating that the students were in the satisfactory–good range ($M = 3.31$) with some variance.

The normality of the dataset and variable scores was examined using standard procedures for exploring distribution characteristics. Normality values ranged from -0.87 to 0.04 for skewness and from -0.98 to 0.07 for kurtosis for all variables, which were well within acceptable ranges (Kline, 2011). Shapiro–Wilk’s test (Cramer, 1998) of the achievement data demonstrated that the GPA scale was approximately normally distributed with a skewness of -0.21 ($SE = 0.17$) and a kurtosis of -0.44 ($SE = 0.35$).

Structural Equation Modeling

SEM with Mplus 6 was used to verify the proposed model and address the research questions. Based on Muis’ (2007) assumptions, two models were constructed: an

Table 1 Descriptive statistics for the variables

Measure and subscale	<i>M</i>	<i>SD</i>	Skewness	Kurtosis
MEBQ^a				
Justification	3.21	1.11	−0.87	0.12
Certainty	3.62	1.31	−0.73	−0.61
Source	3.84	1.23	−0.21	−0.81
Development	3.71	1.08	−0.76	0.14
MSLQ^b				
Intrinsic motivation	4.72	1.31	−0.64	0.07
Extrinsic motivation	5.21	1.71	−0.43	−0.88
Rehearsal	5.24	1.13	−0.77	0.47
Elaboration	4.71	1.87	−0.48	−0.73
Organization	5.13	1.73	−0.46	−0.60
Critical thinking	4.94	1.89	−0.58	−0.73
Metacognitive self-regulation	4.01	1.86	−0.04	−0.98
GPA ^a	3.31	1.10	−0.21	−0.61

M mean, *SD* standard deviation, *MEBQ* modified epistemic beliefs questionnaire, *MSLQ* motivated strategies for learning questionnaire, *GPA* grade point average

^a 1–5-point scale

^b 1–7-point scale

original model based on Muis' assumptions (Fig. 2a) and the competing model (Fig. 2b) based on additions to Muis' assumptions that tests if personal epistemology directly predicts cognitive and metacognitive strategies. The rationale for constructing the competing model was to find the best-fitting model with data. Moreover, previous studies reported that students' ideas about knowledge and knowing were directly related to their cognitive and metacognitive strategies (Paulsen & Feldman, 2007). In Fig. 2a, the solid line between personal epistemology and goal orientation of SRL corresponds to propositions 1 and 2 in Muis' (2007) model—personal epistemology is one component of the cognition and affective conditions on task definition, which influences goal standards that students set. In Fig. 2a, the dashed line between personal epistemology and cognitive and metacognitive strategies refers to proposition 3—personal epistemology translates into epistemic standards that serve as inputs to metacognition.

The competing model yielded better-fit indices and explained more variances of cognitive and metacognitive strategies than the original model. Table 2 displays the fit indices of the original and competing models after putting in the same intercorrelations based on the modification indices. The chi-square difference test yielded significant difference between the original and the competing models, $\chi^2_{\text{dif}}(20) = 38.59, p < .001$ (Kline, 2011). Furthermore, the Akaike information criterion (AIC) and Bayesian information criterion (BIC) values were lower for the competing model than the original model. Kline (2011) suggested that lower AIC and BIC values and better-fit indices were justification for the interpretation favoring the competing model (Fig. 2b) over the original proposed model (Fig. 2a).

A series of analyses were conducted. The first run of the competing model (Fig. 2b) yielded a poor fit of the data, $\chi^2(17, N = 209) = 206.39, p < .001$, SRMR = 0.09, RMSEA = 0.10, CFI = 0.73. Then, the modification indices of the competing model were examined. Based on those indices, the intercorrelations between (a) intrinsic and

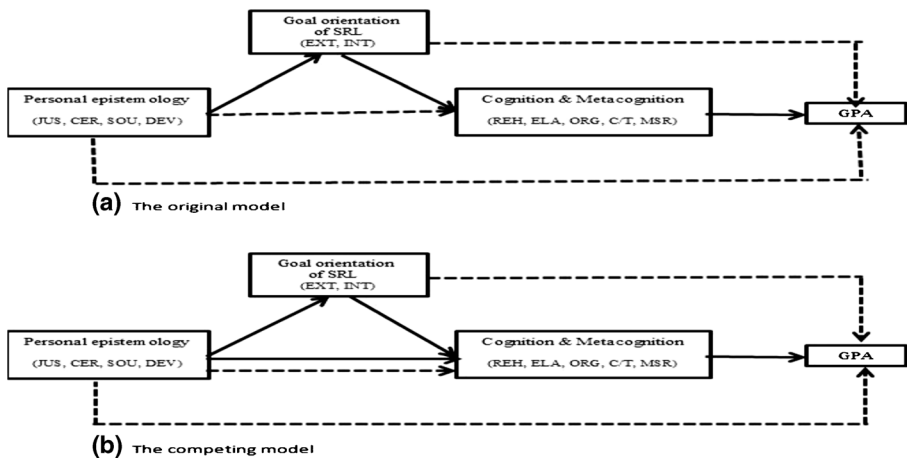


Fig. 2 a Represents the original model based on Fig. 1 and b represents the competing model. Solid lines are for direct relationships, and dashed lines are for indirect relationships. JUS justification of knowledge, CER certainty of knowledge, SOU source of knowledge, DEV development of knowledge, EXT extrinsic motivation, INT intrinsic motivation, REH rehearsal, ELA elaboration, ORG organization, C/T critical thinking, MSR metacognitive self-regulation

Table 2 Fit indices of the original and competing models

Model	χ^2	<i>df</i>	CFI	AIC	BIC	RMSEA
Original	62.13	32	0.95	5568.5	5782.4	0.06
Competing	23.54	12	0.98	5567.0	5714.2	0.05
χ^2 test	$\chi^2_d=38.59$	$df_d=20$	$p < .001$			

CFI comparative fit index, AIC Akaike information criterion, BIC Bayesian information criterion, RMSEA root mean square error of approximation

extrinsic motivations, (b) rehearsal and metacognitive self-regulation, (c) rehearsal and organization, (d) elaboration and critical thinking, and (e) organization and metacognitive self-regulation were recognized. These correlations were also conceptually reasonable; for example, students may organize information to use in a rehearsal strategy. These intercorrelations are shown with curved dashed lines in Fig. 3 (for graphical simplicity, only significant estimates are shown).

Overall, the fit statistics revealed that the competing model fit the data collected in this study quite well. More specifically, the chi-square statistic, χ^2 (12, $N = 209$) = 23.54, was significant ($p < .03$), indicating that although the model did not fit the data well, the chi-square/*df* ratio smaller than 2 indicated an adequate fit (Kline, 2011). Based on Hu and Bentler's (1999) criteria, the indices of the competing model were a good fit, χ^2 (12, $N = 209$) = 23.54, $p < .01$, SRMR = 0.03, RMSEA = 0.05, CFI = 0.98. Coefficients for the direct causal pathways were calculated and are presented in Fig. 3.

The model explained 11.6 % of the variance in students' physics achievement. This can be considered a good result because it can be accounted for by the students' physics-related personal epistemologies and SRL strategies. Furthermore, the model

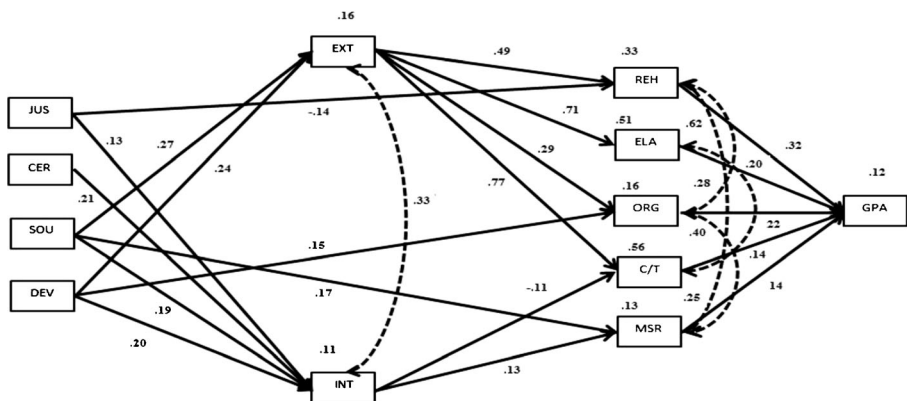


Fig. 3 Solid lines denote direct relationships and curved dashed lines are for the paths that were added after examining modification indices. The number above each observed variable represents variance explained by the model. JUS justification of knowledge, CER certainty of knowledge, SOU source of knowledge, DEV development of knowledge, EXT extrinsic motivation, INT intrinsic motivation, REH rehearsal, ELA elaboration, ORG organization, C/T critical thinking, MSR metacognitive self-regulation

explained the other variances: 16 % in extrinsic goal orientation, 11 % in intrinsic goal orientation, 33 % in rehearsal, 51 % in elaboration, 14 % in organization, 57 % in critical thinking, and 13 % in metacognitive self-regulation.

Examination of the path coefficients revealed that some dimensions of personal epistemology were significantly related to some dimensions of the MSLQ, yet others were not. Two dimensions describing the process of knowing were positively related to the SRL motivation dimensions; source of knowledge explained significant variance in extrinsic motivation ($\beta = 0.27, p < .001$) and intrinsic motivation ($\beta = 0.19, p < .001$). Similarly, beliefs in justification of knowledge predicted students' intrinsic goal orientation ($\beta = 0.13, p < .05$); yet it was not significantly related to extrinsic motivation ($\beta = 0.04, p = .41$). The other two dimensions concerning the nature of knowledge were also positively associated with the SRL motivation dimensions. Beliefs in the development of knowledge accounted for significant ($p < .001$) variance in extrinsic motivation ($\beta = 0.24$) and intrinsic motivation ($\beta = 0.20$). The certainty of knowledge explained significant ($p < .001$) variance in intrinsic motivation ($\beta = 0.21$); however, certainty did not account for significant ($p = 0.39$) variance in extrinsic motivation ($\beta = 0.04$).

Some dimensions of personal epistemology were directly associated with some dimensions of cognitive and metacognitive strategies. Beliefs in justification of knowledge were significant ($p < .01$) and inversely related to rehearsal strategies ($\beta = -0.14$), indicating that the more sophisticated belief in justification was related to less often use of rehearsal strategies. The source of knowledge was directly and positively related to metacognitive self-regulation ($\beta = 0.17, p < .001$). The direct relationship between development of knowledge and organization strategies was significant ($\beta = 0.15, p < .001$).

Additionally, intrinsic goal orientation significantly predicts use of critical thinking ($\beta = -0.11, p < .05$) and metacognitive self-regulation ($\beta = 0.13, p < .01$). Extrinsic goal orientation was significantly ($p < .01$) related to use of rehearsal ($\beta = 0.49$), elaboration ($\beta = 0.71$), organization ($\beta = 0.29$), and critical thinking ($\beta = 0.77$). Furthermore, the intercorrelation based on the modification indices indicated that extrinsic goal orientation was positively related to intrinsic goal orientation ($\beta = 0.33, p < .01$).

Another relationship revealed by the analysis was the positive significant ($p < .001$) effect of rehearsal strategies on students' academic achievement in physics ($\beta = 0.32$), which suggests that students who reported more often use of rehearsal strategies to learn physics received higher scores in the semester tests. Likewise, organization strategies accounted for significant ($p < .001$) variance in academic achievement ($\beta = 0.22$) indicating that the more frequently students organized the information during a particular learning task, the higher their GPA at the end of the semester. In addition, elaboration ($\beta = 0.20, p < .001$), critical thinking ($\beta = 0.14, p < .01$), and metacognitive self-regulation strategies ($\beta = 0.14, p < .01$) significantly predicted academic achievement.

Examination of the total indirect effect of personal epistemology on SRL strategies through the mediation of motivation yielded significant effects. Table 3 summarizes the coefficients for the indirect relationships among epistemic beliefs, self-regulated strategies, and physics achievement. The mediation of extrinsic goal orientation beliefs on source and development of knowledge accounted for significant variance in rehearsal

($\beta = 0.18$ and $\beta = 0.11$, $p < .01$), elaboration ($\beta = 0.26$ and $\beta = 0.24$, $p < .01$), and organization ($\beta = 0.12$ and $\beta = 0.11$, $p < .05$) strategies, and critical thinking ($\beta = 0.30$ and $\beta = 0.27$, $p < .01$), respectively. Additionally, source of knowledge indirectly predicted metacognitive self-regulation strategies through the mediating role of extrinsic goal orientation ($\beta = 0.07$, $p < .05$). However, beliefs in justification and certainty of knowledge did not indirectly predict students' reported usage of cognitive and metacognitive strategies in physics.

Another indirect relationship revealed was that students' ideas about the nature of knowing physics significantly predicted their physics achievement; yet, ideas about the nature of knowledge did not. Beliefs justification ($\beta = 0.07$, $p < .01$) and source of knowledge ($\beta = 0.09$, $p < .01$) significantly and indirectly explained the variance in physics achievement through the mediating role of metacognitive self-regulation. Similarly, extrinsic goal orientation indirectly predicted physics achievement through mediation of cognitive and metacognitive strategies ($\beta = 0.15$, $p < .01$).

Conclusion and Implications

The recent emphasis in science education research is one of the important *hot* constructs on learning science (Sinatra, 2005). Therefore, not only do the results of this study provide evidence for Muis' (2007) theoretical model and hypothesis but they also give insight into how some hot constructs, including personal epistemology and SRL, interact with and impact the ways of knowing physics and achievement scores in physics. The results of this study indicate that both students' views on the nature of knowledge and knowing in physics positively predicted their goal orientations. More specifically, in terms of the nature of knowledge in physics and consistent with Muis and Franco (2009), we found that students who viewed physics knowledge as tentative ($\beta = 0.21$) and complex ($\beta = 0.20$) adapted intrinsic goal orientations toward learning physics. Furthermore, we found that students who believed evidence is required to evaluate the physics claims ($\beta = 0.13$) and knowledge is internally constructed

Table 3 Indirect effects from personal epistemology to cognitive strategies and physics achievement

	REH		ELA		ORG		C/T		MSR		GPA
	EXT	INT	EXT	INT	EXT	INT	EXT	INT	EXT	INT	
JUS	0.02	0.01	0.05	0.01	0.02	0.01	0.06	-0.01	0.01	0.02	0.07*
CER	0.03	0.01	0.06	0.01	0.03	0.01	0.06	-0.02	0.01	0.04	-0.02
SOU	0.18*	0.01	0.26*	0.01	0.12*	0.01	0.30*	-0.02	0.07**	0.04	0.09*
DEV	0.11*	0.01	0.24*	0.01	0.11*	0.01	0.27*	-0.02	0.06	0.04	0.04
INT	—	—	—	—	—	—	—	—	—	—	0.01
EXT	—	—	—	—	—	—	—	—	—	—	0.15*

REH rehearsal, ELA elaboration, ORG organization, C/T critical thinking, MSR metacognitive self-regulation, GPA grade point average, EXT extrinsic motivation, INT intrinsic motivation, JUS justification of knowledge, CER certainty of knowledge, SOU source of knowledge, DEV development of knowledge

* $p < .01$; ** $p < .05$

($\beta = 0.19$) were more likely to possess intrinsic goal orientations toward learning physics. However, students who viewed the knowledge as internally constructed ($\beta = 0.27$) possessed extrinsic motivation toward learning physics. These findings are important because recent studies in physics education highlighted the positive effects of goal orientations in conceptual change (Taasoobshirazi & Sinatra, 2011) and in attitudes (Gungor, Eryilmaz & Fakioglu, 2007) in learning physics. Taasoobshirazi and Sinatra (2011) suggested students should be encouraged to have a combination of intrinsic and extrinsic goal orientations since both support conceptual change and learning in physics. The positive influences of personal epistemology on fostering a combination of intrinsic and extrinsic goal orientations were supported by the findings of this study.

Duit, Neidderer and Schecker (2007) claimed that what makes learning physics so difficult for students is its extremely high levels of abstraction and idealization. Thomas (2013) underscored the importance of metacognitive strategies in learning physics that requires students to revise both what it means to learn physics and their physics learning process. The findings of this study give important insight into the relationships among students' personal epistemology and cognitive and metacognitive strategies in physics. The results indicate that personal epistemology was indirectly related to cognitive and metacognitive strategies through goal orientations. More specifically, we found that the students who viewed knowledge as internally constructed reported more frequent use of elaboration strategies. Similarly, the use of critical thinking strategies was indirectly predicted by beliefs that physics knowledge is evolving. Both of these findings reflect the dynamic and tentative nature of physics and the need for in-depth and enriched understanding.

The findings of this study can have practical implications for physics teaching and curriculum developers in Turkey. One important finding is that source of knowledge was positively related to intrinsic goal orientation, metacognitive strategies, and achievement. The positive correlation indicated the more sophisticated belief in source was related to greater levels of intrinsic motivation, which led students to study physics for curiosity and to develop metacognitive strategies and skills, which helped them monitor and regulate their learning and achieve higher course grades in physics. Sin (2014) argued that in physics classes traditional teaching activities were focused on acquisition of certain knowledge that ignores the process of knowledge production. Furthermore, previous studies documented that typically students describe their laboratory activities as simple and highly structured (Ibrahim, Buffler & Lubben, 2009) and traditional physics instruction tells students exactly what is needed to be done (Samarapungavan, Westby & Bodner, 2006). This sort of traditional teaching in physics may lead students to develop an idea that knowledge in physics comes from authority and is certain (Topçu, 2013). Such teaching and related examination requirements might not encourage the full development and engagement of students' curiosity, self-regulated learning, and accurate views of the nature of physics.

In response to this problem, many scholars recommend that science teaching activities should reflect epistemological aspects of authentic inquiry experiences (Chinn & Malhotra, 2002; Sin, 2014). Koponen and Mantyla (2006) suggested that the generative justification of knowledge, drawing insight from history and philosophy of physics, is based on inductive generalizations. Rather than re-enacting historical

experiments at school, it emphasizes considering their epistemology, where experiments are the source of new knowledge. When students begin to understand the epistemological aspects of the experiments, they will be a better judge of the ways to approach experiments in physics and may develop an idea that knowledge is constructed by the learner (Chinn & Malhotra, 2002; Koponen & Mantyla, 2006).

Another important finding of this study is the negative relationship between justification of knowledge and the usage of rehearsal strategies. Previous studies reported that traditional physics instruction at schools is mostly about teaching how to use formulas in physics problems (Meltzer, 2002; Redish & Steinberg, 1999). Students describe that mastering physics includes knowing how to use formulas in physics problems (Redish & Steinberg, 1999). Topçu (2013), for example, found that passive reproduction was well engrained in the education system; he reported that during physics courses Turkish preservice physics teachers only listened to their instructors and took notes. This sort of teaching instruction might have encouraged students who were extrinsically motivated to choose rehearsal strategies more often in order to memorize information and procedures to perform better at physics examinations consisting of recall questions and problem solving. However, in this study, it was found that the more sophisticated belief that knowledge in physics comes from experimenting and reasoning (i.e. justification of knowledge) led the students to use fewer rehearsal strategies. Fostering sophisticated beliefs about justification of knowledge may be beneficial to help students use less memorization and encourage them to critique knowledge in physics.

Physics teaching needs to emphasize argument, critique, and explanation to develop more sophisticated beliefs in justification of knowledge since physics is as much about argument as inquiry. Argumentation is a social practice grounded in particular frameworks about what counts as claims, data, justification, and evidence. Therefore, argumentation and critique can promote satisfactory changes in how students view and use reasoning and experimenting in justifying scientific knowledge claims in physics (Dole & Sinatra, 1998). Research has documented that students seem to struggle with argumentation in classroom practices (Ryu & Sandoval, 2012). However, some scholars have suggested scaffolding as a way to develop students' argumentation skills, metacognition, and SRL skills (Lin, Hsu, Lin, Changlai, Yang & Lai, 2012).

An underlying assumption of the theoretical model used to focus the design, data collection, and analysis of this study is that the relationship between personal epistemology and SRL is reciprocal and domain specific. The structure of physics, for example, is perceived by students as well defined and includes many technical terms that are tightly structured (Muis et al., 2006). Consistent with this view, previous studies on students' ideas about learning physics reported that learning physics means effectively memorizing how to use formulas (Ehrlich, 2002; Sin, 2014); such views and strategies are frequently reinforced by assessment methods and how students define achievement in physics. Studies reported that some students believed that "studying historical and real-life examples will not be rewarded, and formulas will appear on the test" (Elby, 1999, p. 53). Topçu (2013) argued that the only concern of the physics teacher is to test the students' content knowledge. Inevitably, rehearsal was the most often used strategy by the students in this study ($M = 5.2$). However, frequent use of such surface-processing strategies discourages students from attaining conceptual understanding in physics and fails them in constructivist-oriented assessments. Yumusak

et al. (2007) reported a negative correlation between rehearsal strategies and achievement in biology assessed by a constructivist-oriented test. We used students' final course grades that included the grades from three teacher-made examinations, which likely stressed recall of information and traditional problem-solving questions. This result suggests that physics teachers should be encouraged and supported in the development and use of constructivist-oriented or alternative assessments that measure students' physics learning from a broader perspective, not just their memorizing-formula skills.

One assumption in Muis' (2007) model is that personal epistemology is indirectly related to students' usage of cognitive and metacognitive strategies through the mediation of motivation. However, the results of this study demonstrated that students' ideas about knowledge and knowing directly predicted their cognitive and metacognitive strategies. Consistent with this study's results, Paulsen and Feldman (2007) reported that personal epistemology successfully accounted for ~16 % of the variance in cognitive and metacognitive strategies ($p < .05$) and that beliefs on the development of knowledge were correlated with elaboration ($\beta = 0.18$) and rehearsal strategies ($\beta = -0.21$). Bromme et al. (2010) discussed that personal epistemology directly influences any phase of SRL by influencing students' internal standards for metacognitive monitoring and regulating. Although the results of this study provide evidence for these claims, there is a need to re-test the model in this study in different contexts.

Dominance of extrinsic motivation over the students' cognitive and metacognitive strategies may be due to a fact that their GPA from the physics course constitutes an important part of their score at the UPE. The Turkish educational system uses 0.6 of the high school GPA (out of 100) added to the UPE score to establish the university entrance score. Therefore, the students in this study may be more extrinsically motivated to obtain good grades. Consequently, their cognitive and metacognitive strategies may be based on what strategy works better for achieving good grades. A limitation of this study was that the data were collected in May when students were about to take their final examination in the physics course and examination pressures were at their highest. Any attempt to generalize the findings of this study should be approached with caution.

This study has other limitations. First, the model explained only 12 % of the variance in students' achievement in physics. A large majority of the variance (88 %) remained unexplained; therefore, further studies with additional variables to personal epistemology and SRL (e.g. attitudes, intention, identity, self-efficacy, etc.) are needed to expand the model. Second, the students' physics course grade was taken as their achievement indicator. Because the researcher does not have any control over the preparation of the assessments used, generalization of this study's findings regarding academic achievement in physics should be approached with caution. Third, due to their unavailability at the time the study took place, grade 12 students did not participate in this study; as well, data were collected from only one school. Therefore, generalization should be approached with caution. Lastly, as the study context influences both personal epistemology and SRL, the model may not neatly give the same relations in another context (e.g. university, country). Thus, there is a need for more studies on the relationship between personal epistemology and SRL in physics for the generalizability of the model and its implications.

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